

CentOS Firewall Configuration Guide

A simple step-by-step documentation for configuring `firewalld` on CentOS

Step-by-Step Configuration Steps

STEP 1

Check Firewall Status

Verify if the firewall service is actively running on your CentOS system:

```
sudo firewall-cmd --state
```

Expected Output: running

STEP 2

Allow Nginx Web Traffic (Port 80)

Open Port 80 permanently to allow incoming web traffic through the firewall:

```
sudo firewall-cmd --permanent --add-service=http
```

Alternative Port Command: `sudo firewall-cmd --permanent --add-port=80/tcp`

STEP 3

Reload Firewall Rules

Apply the permanent changes without interrupting existing active connections:

```
sudo firewall-cmd --reload
```

Expected Output: success

STEP 4

Verify Allowed Services & Ports

Confirm that http or 80/tcp is listed under active allowed rules:

```
sudo firewall-cmd --list-all
```

Command Quick Reference Summary

Task	CentOS Command	Purpose
Check Status	sudo firewall-cmd --state	Confirms firewall is active.
Allow Web Service	sudo firewall-cmd --permanent --add-service=http	Opens HTTP port 80 permanently.
Reload Rules	sudo firewall-cmd --reload	Applies saved configuration.
List All Rules	sudo firewall-cmd --list-all	Displays active firewall configuration.

Key Takeaway for Documentation: The --permanent flag saves the rule across system reboots, while sudo firewall-cmd --reload forces the firewall to apply those changes immediately.